Software Requirements Specification

HASH Project — MVP Summary

Maternal Care & Early Pregnancy Loss Support Platform

Track I — Early Pregnancy Loss Care

Document Version: MVP 1.2 (Choronko/GSM + Post-Loss Care as MVP)

Status: Working Document for the Team

1. Overview

1.1 What We're Building

HASH is a maternal-care platform that follows pregnant women through their pregnancy and supports them after a pregnancy loss. Patients talk to an AI chatbot. Hospitals see those conversations and any emergencies on a dashboard.

The MVP delivers five things end-to-end:

1. Hospital and patient onboarding on two tracks: smartphone (app/web sign-up) and choronko (clinician-onboarded SMS user, phone number as unique identifier). Includes a baseline patient risk classification (Low / Medium / High) computed from the clinical questionnaire.
2. A chatbot that does wellness check-ins (at a frequency based on the patient's risk level), gestational tips, and reminders — delivered through whichever channel the patient is on.
3. AI message triage that classifies each individual reply as Low, Medium, or High acuity and reacts accordingly — including alerting the hospital for High.
4. Post-pregnancy-loss care: a first-class care track that activates when a clinician marks a loss. Gentle conversational check-ins, mental-health watch (plain-language PHQ-2), a crisis route for self-harm signals, and immediate response to opt-out keywords. Works on both smartphone and choronko.
5. A hospital dashboard that shows patients (with risk level and track), emergencies (with GPS for smartphone patients), appointments, and a dedicated post-loss view.

Two important concepts to keep separate from the start:

- Patient risk level — set at onboarding from clinical history (age, parity, comorbidities, previous loss, etc.). Decides how often the bot proactively checks in on her. Changes only when a clinician revises it.
- Message acuity — the priority of any single message a patient sends. Decides the bot's immediate response and whether to alert the hospital. Per-message, real-time.

These two run in parallel. A Low-risk patient can still send a High-acuity message and trigger an emergency alert. The bot will reduce her check-in frequency back to fortnightly afterwards, unless a clinician decides to raise her risk level.

1.2 Who Uses It

User	Where they use it	What they do
Smartphone patient (pregnant or post-loss)	Mobile/Web app + SMS fallback	Chat with the bot, see tips and reminders, press emergency button when needed.

User	Where they use it	What they do
Choronko patient (feature phone / SMS only)	SMS	Same care as smartphone patient — receives check-ins, tips, reminders, and crisis support. Reports symptoms by text.
Hospital personnel	Web dashboard	See assigned patients (both tracks), respond to emergency signals, view appointments and patient condition graph.
Clinician (doctor/midwife/nurse)	Web dashboard	Onboard choronko patients, review patient conversations, add notes, mark cases (including pregnancy loss), revise risk levels.

1.3 Channels in the MVP

- Mobile App / Web App: full experience for smartphone patients — chat, home-screen flash cards, EDD countdown, emergency button with GPS, push notifications.
- SMS: the full experience for choronko (feature-phone) patients, who use the platform entirely via SMS. The same patient may also receive SMS as a fallback when she's a smartphone user with the app closed.
- WhatsApp: planned for Phase 2.

1.4 Hard Rules (Non-negotiable)

- The bot never prescribes drugs or dosages.
- Every clinical message the bot can send is approved by Dr Elvira (clinical lead) before release.
- Patient data is encrypted, access-logged, and consent-based.
- Pregnancy-loss content is gentle, opt-out-friendly, and never auto-triggered — only after a clinician marks the case.

2. MVP Features (Detailed)

2.1 Onboarding

2.1.1 Hospital Onboarding

Sign-up captures:

- Hospital name.
- GPS location.
- Physical address.
- Hospital contact info (phone, email).
- Password (for the hospital admin account).
- Personnel name and contact (the first staff user).

After sign-up, the hospital admin can add additional personnel and edit the hospital profile.

2.1.2 Patient Onboarding

Sign-up captures:

- Name.
- Phone number — this is the patient's unique identifier across the platform, including the choronko (SMS-only) track.
- Age.
- Password (smartphone track only).
- Clinical sign-up questions (final list to be provided by Dr Elvira — placeholder fields: LMP/EDD, parity, known comorbidities, previous loss).

Two patient tracks: smartphone and choronko. Both are first-class — neither is a stripped-down version of the other.

- Smartphone track: patient signs up in the mobile or web app, chats with the bot in-app, receives push notifications, sees flash cards on the home screen, can press the emergency button.
- Choronko track (feature phone / GSM): patient does not own a smartphone. A clinician onboards her at the hospital and records her phone number. That number becomes her unique identifier for everything that follows. She interacts with the bot entirely via SMS — no app, no web.

A choronko patient is not a second-class user. She still receives:

- Wellness check-ins at the cadence set by her risk level (daily / weekly / fortnightly).
- Gestational tips matched to her current week — same content as smartphone users, formatted for SMS (short, plain text, no markdown).

- Practical reminders and small advisories ("Remember to drink water today," "Try to rest after lunch," "Eat iron-rich food this week").
- Appointment reminders by SMS.
- The full message-triage pipeline on her replies — if she texts back "I am bleeding," the hospital is alerted exactly as it would be for a smartphone patient.
- Post-loss support over SMS (see section 2.7).

What the choronko track can't offer (without a smartphone): in-app countdown, push notifications, flash cards, and the GPS emergency button. For emergencies, the hospital uses the patient's registered address and her own text-message report. If she later acquires a smartphone, she can claim her account using her phone number and continue with full functionality.

2.1.3 Patient Risk Classification (at Onboarding)

At the end of the sign-up clinical questionnaire, the system computes a baseline risk level for the patient — Low, Medium, or High. This is a patient-level attribute, set once at onboarding and updated only when the clinician revises it. It is different from the per-message triage in section 2.3.1.

The risk level is derived from the answers Dr Elvira's questionnaire collects, for example:

- Age outside the typical range (e.g. ≤17 or ≥35).
- Parity and obstetric history (previous loss, previous stillbirth, previous caesarean, previous pre-eclampsia).
- Comorbidities (hypertension, diabetes, sickle-cell, HIV, severe anaemia).
- Multiple pregnancy.
- Late ANC initiation, or no prior antenatal care.

The exact scoring rubric (which factors count, with what weight, and what the Low/Medium/High thresholds are) is defined by Dr Elvira. The system stores both the computed risk level and the answers that produced it, so the clinical lead can revise the rubric over time without losing history.

The patient's risk level drives:

- How often the bot performs wellness check-ins (section 2.3.2).
- The starting band on the dashboard condition graph (section 2.6).
- Which clinician on the hospital side is suggested for review of the patient (Phase 2).

A clinician can change a patient's risk level at any time from the dashboard — for example, after a hospital visit reveals new information. Changes are logged and the check-in cadence updates immediately.

Important distinction: a Low-risk patient can still send a High-priority message (e.g. heavy bleeding) — and the system will alert the hospital exactly the same way. The two concepts are independent and both run in parallel.

2.2 Dashboard (Hospital)

- Lists all patients assigned to the hospital.
- Shows emergency signals from patients (with GPS location when triggered by the emergency button).
- Shows upcoming patient appointments.
- Per-patient view with a condition graph that moves from green → yellow → red across gestation.
- Important dates for each patient (appointments and clinically significant pregnancy dates).

2.3 Conversational Follow-Up

Two independent mechanisms run here. Don't confuse them:

- Message triage (2.3.1) — classifies each individual message a patient sends. Per-message, real-time.
- Wellness check-ins (2.3.2) — outbound questions the bot asks on a schedule. Frequency is determined by the patient's onboarding risk level (2.1.3), not by any single message.

2.3.1 Message Triage and Non-Prescriptive Guidance

When a patient sends a message, the system classifies that specific message into Low, Medium, or High acuity and responds accordingly. This classification is per-message and does not change the patient's risk level.

Message acuity	Bot response	What happens next
Low	Reassuring message and self-care tip.	Logged to dashboard. No alert.
Medium	Empathic reply and a short set of follow-up questions to gather more information.	Schedule a re-check; flag the case on dashboard for clinician awareness.
High	Urgent guidance — go to hospital, basic first-aid steps, no prescription.	Hospital alerted in real time (see 2.5). GPS sent if the patient used the emergency button.

Note: a High-acuity message from a Low-risk patient is treated as a High-priority alert — message acuity always wins in the moment. If a patient repeatedly sends Medium- or High-acuity messages over a short window, the clinician may decide to escalate her overall risk level on the dashboard.

2.3.2 Wellness Check-ins

Outbound questions the bot proactively asks the patient to confirm she is okay. Frequency is set by the patient's risk level from onboarding (section 2.1.3), not by any single message:

Patient risk level	Check-in frequency	Example question
High-risk patient	Daily	"Good morning. Any bleeding, severe headache, or unusual pain today?"
Medium-risk patient	Weekly	"How have you been feeling this week? Any new symptoms?"
Low-risk patient	Fortnightly (to be confirmed by Dr Elvira), with extra check-ins around key gestational milestones	"Hi! Just checking in. Anything you'd like to share about how you're feeling?"

Each check-in reply goes through the same message-triage pipeline as any other patient message. So a Low-risk patient's reply to a fortnightly check-in can still raise a High-acuity alert if she reports something dangerous.

If a patient consistently misses check-ins (e.g. three in a row for High-risk or two in a row for Medium-risk), the dashboard flags her for outreach by a clinician.

2.4 Gestational Tips and Appointment Reminders

2.4.1 Gestational Tips

- Pregnancy tips matched to the patient's current gestational week and her recorded health information.
- Post-loss tips on physical recovery and mental-health self-care after a miscarriage.
- Delivery format is flexible: tips can be shown as in-app notifications, SMS messages, in-conversation messages from the bot, or as flash cards on the app home screen.

2.4.2 Appointment Reminders

- Triggered by appointments the patient schedules in the app.
- Delivered as notifications, SMS, and in-conversation reminders from the bot.

2.4.3 Estimated Time of Delivery (EDD)

- Calculated from LMP and any other relevant factors the user provides.
- Displayed on the home screen as a countdown to delivery.
- Raises awareness near term: from about one month before EDD, the app increases reminders so the patient takes extra care and keeps a small bag with her in case her water breaks.

2.5 Emergency Button & GPS

- Visible button in the app for emergencies.
- When pressed, the system captures the patient's GPS location and sends an alert to her assigned hospital.

- Alert payload includes the patient's recent conversation context so the hospital knows what's happening.
- The hospital can use the location to dispatch help to the patient or coordinate transport.

2.6 Patient Condition Graph

- Per-patient visualisation on the dashboard.
- Plots the patient's condition over time using her triage history and check-in responses.
- Colour bands: green (stable), yellow (medium concern), red (high concern).
- Important pregnancy dates and appointments are marked on the timeline.

2.7 Post-Pregnancy-Loss Care

Post-loss care is a first-class part of the MVP, not an afterthought. It is the most sensitive part of the platform and is designed with emotional intelligence as the leading principle, not the last.

Activation. The post-loss track only begins when a clinician marks the pregnancy as Loss on the dashboard, recording the date of loss and the gestational age. The system never infers a loss from chat messages alone, no matter what the bot thinks it heard.

Immediate effects on activation:

- All routine pregnancy reminders, tips, and the EDD countdown stop within 5 minutes.
- The chatbot switches to the post-loss content set (separate from the pregnancy content set in M5).
- The first outbound message is the opener approved by Dr Elvira — gentle, acknowledges the loss, offers support without pushing it, and gives the patient a clear way to pause or stop messages.
- Available on both tracks: smartphone patients see the change in the app; choronko patients receive the same care entirely over SMS, in shorter, plain-text form.

2.7.1 The Three Things the Bot Does Post-Loss

1. Listen. The bot's main job here is not to advise — it's to be there.

- Asks gentle, open check-in questions instead of clinical ones ("How are you doing today?" not "Rate your symptoms.").
- If she replies, the bot acknowledges what she said in plain, warm language before doing anything else. No clinical interrupts mid-sentence.
- If she doesn't reply, the bot does not nag. After two missed check-ins, the dashboard quietly flags her for a clinician to consider reaching out — it doesn't push the patient harder.
- Messages from the bot use her name, never generic templates, and never repeat the words "miscarriage" or "loss" more than necessary.

2. Check on mental health — without sounding like a survey.

- The bot watches for signs of distress (sadness lasting weeks, sleep disturbance, loss of appetite, hopelessness, anxiety).
- MVP scope: a short, conversational two-question check (adapted from PHQ-2 in plain language) offered at week 2 — not a clinical instrument and not framed as one.
- If the patient declines or doesn't engage, the bot doesn't escalate the check — it simply notes "patient did not engage with mental-health check this week" on the dashboard.
- Phase 2 adds the longer PHQ-9 and EPDS at weeks 4, 8, and 12. The MVP only ships the gentle PHQ-2 version.

3. Catch emergencies (physical and emotional).

- Physical red flags (heavy bleeding, foul-smelling discharge, fever >38°C, severe pain): treated as High-acuity messages, alert the hospital exactly as in section 2.3.1.
- Mental-health crisis signals ("I want to die," "I don't want to live," "hurt myself," or similar): immediately High acuity, alert the hospital, and send a crisis-resource message with local hotline numbers — also approved in advance by Dr Elvira.
- The crisis-resource message goes out by SMS even to smartphone patients, so it lands even if the app is closed.

2.7.2 Emotionally Intelligent Defaults

These are not nice-to-haves. They are the difference between care and harm.

- Tone, not just words. Replies are short, warm, and unhurried. No exclamation marks, no emojis unless the patient uses them first, no platitudes ("everything happens for a reason"). All checked into the M5 content library by Dr Elvira.
- Pacing. After activation, only the opener is sent on day one. The next message comes 48 hours later, then every few days, settling into a weekly check-in by week three. Cadence is overridden by the patient's risk level for physical recovery (a High-physical-risk post-loss patient is still checked on daily for the first week).
- Opt-out is one keyword. PAUSE silences messages for a week. STOP silences them indefinitely. RESUME brings them back whenever she's ready. Either keyword gets a single confirmation message — no follow-up nudges.
- No automatic talk of future pregnancies. The bot never volunteers contraception advice or talk of "trying again" in the MVP. If the patient asks, the bot offers a referral conversation, scheduled through a clinician.
- Faith, family, and culture. Where the patient indicated a preference at onboarding (e.g. faith-based support or peer-support groups), the bot offers those resources — only as opt-in, never as default.

- No bot judgement on cause. The bot never speculates about why the loss happened and never reframes the loss for the patient ("it was God's will," "it's for the best" — banned phrasings in the content review).

2.7.3 What the Dashboard Shows

- A clear post-loss banner on the patient's page so any clinician opening her record knows the context before reading anything else.
- Physical recovery timeline (bleeding resolution, return of menses, any infection signs reported).
- Mental-health engagement (did she respond to the gentle check, what did she say, has she gone quiet).
- All alerts (physical or crisis) in the same alert queue as pregnancy alerts, marked as post-loss.
- A button for the clinician to schedule a follow-up call or in-person visit, which triggers an SMS or in-app reminder to the patient at her preferred time.

3. Module Map

The MVP is built as nine focused modules. Anything not in this list is Phase 2 or later.

#	Module	Owner
M1	Identity & Onboarding (hospital + smartphone patient + choronko/GSM)	Backend
M2	Patient & Pregnancy Profile (LMP, EDD, risk level, comorbidities, gestation timeline)	Backend, Doctor
M3	Conversation Engine (chat sessions, risk-based check-in cadence, tips scheduler)	Backend
M4	Message Triage (per-message Low/Medium/High acuity, red-flag rules)	Backend, Doctor, Data Analyst
M5	Clinical Content (pregnancy tips, response templates, post-loss content set)	Doctor, Backend
M6	Emergency & Alerting (real-time alert to hospital + GPS)	Backend
M7	Patient App (Mobile / Web with chat, countdown, emergency button, flash cards)	Frontend
M8	Hospital Dashboard (patient list, alerts, appointments, condition graph, risk controls)	Frontend
M9	Post-Pregnancy-Loss Care (track switch, gentle check-ins, mental-health watch, crisis route)	Doctor, Backend, Frontend

Two cross-cutting concerns supported by the cloud engineer and backend:

- Infrastructure & DevOps on AWS (M9): hosting, CI/CD, secrets, observability.
- Audit, Consent & Security (M10): encryption, logging, consent capture and withdrawal.

3.1 Module Cards

M1: Identity & Onboarding

What it does. Sign-up, login, and account types for hospital, personnel, and patients on both the smartphone and choronko (SMS-only) tracks.

MVP features

- Hospital sign-up: name, GPS, address, contact, password, first personnel.
- Add additional personnel after sign-up.
- Smartphone patient sign-up: name, phone, age, password, clinical questions (Dr Elvira's list).
- Choronko (GSM) patient onboarding: clinician registers the patient at the hospital, records her phone number as her unique identifier — no password, no app required.
- A choronko patient can later claim a smartphone account using the same phone number, preserving her full history.
- Login with password (smartphone); password reset by phone OTP.

- Account type stored on the Patient record (full | choronko) so downstream services know which channel to use.

Primary owner: Backend Developer

M2: Patient & Pregnancy Profile

What it does. The clinical truth about each patient: who she is, where she is in her pregnancy, what's known about her risk, and her overall risk level.

MVP features

- Store LMP, calculate EDD, roll over gestational age weekly.
- Capture comorbidities, parity, previous loss, language preference.
- Compute the patient's baseline risk level (Low / Medium / High) from Dr Elvira's onboarding questionnaire.
- Store the risk-scoring inputs (the answers) alongside the computed level so the rubric can evolve.
- Allow clinicians to revise the risk level from the dashboard; cadence in M3 updates immediately.
- Track pregnancy outcome (ongoing, live birth, loss) — set by clinician.
- Drive the gestational-week scheduler and the risk-based check-in cadence used by M3.
- Bind each patient to exactly one hospital (set during onboarding).

Primary owner: Backend Developer; Doctor signs off on fields, the risk rubric, and validation

M3: Conversation Engine

What it does. Run every patient conversation: receive messages, hand them to triage, schedule tips and check-ins, send replies.

MVP features

- Multi-turn chat with short-term memory for context.
- Scheduler for gestational tips (by week).
- Wellness check-in scheduler — cadence driven by the patient's risk level from M2 (High = daily, Medium = weekly, Low = fortnightly + milestones).
- Re-evaluates the schedule whenever a clinician changes the patient's risk level.
- Appointment-reminder dispatcher (in-app, SMS, in-conversation).
- Supports PAUSE / STOP / RESUME keywords across channels.
- Calls M4 for every inbound message (including check-in replies); uses the classification to pick the response.

Primary owner: Backend Developer

M4: Message Triage (AI)

What it does. Read each individual message a patient sends and assign Low / Medium / High acuity — and choose which response template to send back. Independent of the patient's overall risk level.

MVP features

- Deterministic red-flag layer for keywords like "bleeding," "severe pain," "no fetal movement," "severe headache," "blurred vision" — always runs first.
- AI/NLU classifier for nuance and intent (Amazon Bedrock or Comprehend).
- Always returns a clinician-approved response_id from M5 — never composes free-form medical advice.
- Logs every decision with rule and model versions for clinical review.
- Conservative tuning — prefers over-triage to under-triage.
- Does not modify the patient's risk level in M2. (A clinician can promote a patient's risk level after seeing a pattern of High-acuity messages.)

Primary owner: Backend Developer + Data Analyst (training/evaluation) + Doctor (rules and gold labels)

M5: Clinical Content Library

What it does. The single, versioned source of every word the bot can say to a patient — tips, replies, first-aid steps, post-loss messages.

MVP features

- Gestational-week tips written by Dr Elvira.
- Response templates for each triage outcome.
- Post-loss content (gentle tone, opt-out friendly).
- Crisis-resource messages with local hotline numbers.
- Versioned and hot-reloadable — content changes don't need a code release.

Primary owner: Doctor (author and approver); Backend (storage and API)

M6: Emergency & Alerting

What it does. When something is High priority — or the patient hits the emergency button — get the hospital's attention immediately.

MVP features

- Emergency button in the app captures GPS and sends an alert.
- High-priority triage outputs also raise an alert.
- Alert appears on the hospital dashboard in real time (≤ 30 seconds).
- Parallel notifications: dashboard push, duty-clinician SMS, email.

- Auto-escalation if no acknowledgement in 10 minutes (Phase 2: escalate to a higher-tier facility).

Primary owner: Backend Developer; Cloud Engineer for the real-time channel

M7: Patient App (Mobile/Web)

What it does. Where the patient actually lives in the product — chat, countdown to EDD, flash-card tips, emergency button, profile.

MVP features

- Sign-up and login (M1).
- Chat with the bot.
- Home screen: countdown to EDD, today's flash-card tip, next appointment.
- Emergency button (sends GPS via M6).
- Schedule appointments (drives M3's reminders).
- Profile and consent settings; opt-out and re-entry.

Primary owner: Frontend Developer

M8: Hospital Dashboard

What it does. Where the hospital sees its patients, emergencies, and appointments — and where clinicians do their work on each case.

MVP features

- Login with MFA.
- Patient list, sortable by risk level and last activity.
- Live emergency-signal pane with GPS, recent conversation excerpt, and acknowledgement workflow.
- Appointment calendar across all patients.
- Per-patient view: condition graph (green → yellow → red), risk level (with control to revise), important dates, conversation history, clinician notes.
- Revising a patient's risk level updates her check-in cadence in M3 in real time; change is logged.
- Mark a pregnancy as Live Birth or Loss (triggers post-loss flow when marked Loss).

Primary owner: Frontend Developer

M9: Post-Pregnancy-Loss Care

What it does. A first-class care track activated when a clinician marks a pregnancy as Loss. Designed for emotional intelligence first — listen, watch for mental-health distress, catch emergencies, never rush the patient.

MVP features

- Activates only when a clinician marks the pregnancy as Loss on the dashboard.
- Pauses all routine reminders, tips, and EDD countdown within 5 minutes.
- Switches M3 to use the post-loss content set in M5 (separate from pregnancy content).
- Sends Dr Elvira's approved opening message and follows the paced cadence (day 1, +48h, then settling weekly).
- Gentle two-question mental-health check (plain-language PHQ-2) at week 2 — opt-in, never coerced. Phase 2 adds PHQ-9 and EPDS at weeks 4, 8, 12.
- Crisis route: self-harm keywords immediately classified High, alert hospital, send crisis-resource SMS with local hotlines.
- Honours PAUSE / STOP / RESUME with one confirmation message and no further nudges.
- Works on both smartphone and choronko (SMS) tracks — content shortened for SMS without losing warmth.
- Post-loss banner and dedicated timeline on the dashboard so clinicians see the context immediately.
- Banned-phrase list enforced at content review (e.g. "it's for the best").

Primary owner: Doctor (clinical design, content, sign-off); Backend (state machine and crisis route); Frontend (dashboard banner and post-loss view)

4. Core Data (MVP)

Only the essentials needed for the MVP. Anything in Phase 2 (referrals, full bereavement screening, FHIR, advanced analytics) is omitted here.

Entity	Key fields
Hospital	id, name, gps_lat, gps_lng, address, contact_phone, contact_email, created_at
Personnel	id, hospital_id, name, phone, email, role, password_hash, mfa_enabled
Patient	id, hospital_id, name, phone (unique identifier across the platform), age, password_hash (null for choronko), language, lmp, edd, parity, comorbidities[], consent_flags, account_type (smartphone | choronko), status, risk_level (low | medium | high), risk_level_set_at, risk_level_set_by, preferred_support (none | faith | peer | counsellor)
RiskAssessment	id, patient_id, computed_at, computed_by (system | clinician_id), inputs (jsonb — the questionnaire answers or clinician override), rubric_version, result_level
Pregnancy	id, patient_id, lmp, edd, current_ga_weeks, outcome (ongoing | live_birth | loss), loss_date, ga_at_loss
Message	id, patient_id, direction (in | out), channel (app | sms | whatsapp), raw_text, message_acuity (low | medium | high), response_id, model_version, rule_version, created_at
CheckIn	id, patient_id, scheduled_for, sent_at, response_message_id, cadence_at_send (low | medium | high), is_post_loss (bool)
Alert	id, patient_id, hospital_id, source (message_triage | emergency_button | missed_checkins | post_loss_crisis), gps_lat, gps_lng, status (new | ack | resolved), created_at, acknowledged_at
Appointment	id, patient_id, hospital_id, datetime, type, notes, reminder_state
PostLossCase	id, patient_id, activated_at, activated_by (clinician_id), opener_sent_at, current_cadence, phq2_offered_at, phq2_response, opt_out_status (active | paused | stopped), notes
Tip	id, ga_week_min, ga_week_max, category (pregnancy | post_loss), text, sms_text, language, version
ResponseTemplate	id, response_id, acuity, context (pregnancy | post_loss), text, sms_text, language, version, approved_by, approved_at, banned_phrase_check_passed
AuditLog	id, actor_id, action, target_entity, target_id, timestamp

5. AWS Setup (MVP)

Lean and standard. We pick HIPAA-eligible services only and sign a BAA with AWS before launch.

Layer	Service
API + WebSocket	Amazon API Gateway (HTTP + WebSocket)
Compute	Amazon ECS Fargate (containerised microservices)
Database	Amazon Aurora PostgreSQL (Multi-AZ)
Cache + session state	Amazon ElastiCache for Redis
File storage	Amazon S3 (KMS-encrypted)
Auth	Amazon Cognito (with MFA for personnel)
AI / NLU	Amazon Bedrock + Amazon Comprehend (custom classifier)
Notifications	Amazon SNS (push), Amazon SES (email), SMS provider for clinicians (Twilio/Termii)
Secrets	AWS Secrets Manager
Observability	Amazon CloudWatch, AWS X-Ray
Security	VPC, AWS WAF, AWS KMS, Amazon GuardDuty
CI/CD	GitHub Actions + AWS CodeBuild/CodeDeploy

5.1 How a Patient Message Flows

6. Patient sends a message from the app (or SMS).
7. API Gateway receives it and pushes it onto a queue.
8. Conversation Engine picks it up, updates context in Redis, calls Triage (M4).
9. Triage runs red-flag rules, then the AI classifier. Output: priority + response_id.
10. Conversation Engine fetches the matching template from the Content Library and replies.
11. If priority is High: Alerting service sends a real-time alert (WebSocket) and an SMS to the duty clinician.
12. Everything is logged: the raw message, the classification (with model and rule versions), the response sent, and the audit trail.

6. Who Does What (MVP)

6.1 Frontend Developer

- M7 Patient App: chat UI, EDD countdown, flash-card tips on home screen, emergency button with GPS capture, appointment scheduling, profile and consent screens.
- M8 Hospital Dashboard: login + MFA UI, patient list (filter by risk level and by smartphone vs choronko), live emergency pane, per-patient view with condition graph (green/yellow/red) and important dates, appointment calendar, clinician-note form, choronko onboarding form at the hospital.
- M9 Post-Loss UI: dashboard post-loss banner, dedicated post-loss timeline view, mental-health engagement summary, control to mark Live Birth or Loss.
- Smartphone post-loss in-app experience: softer visual treatment, EDD countdown removed, gentle tip cards in place of pregnancy tips.
- Shared design system between the two apps.
- Real-time updates via WebSocket subscriptions.

6.2 Backend Developer

- M1 Identity & Onboarding for both tracks (smartphone sign-up, hospital-driven choronko onboarding with phone-number identity, later claim flow).
- M2 Patient & Pregnancy Profile (LMP, EDD calculation, GA roll-over, risk-level computation, scheduler hooks).
- M3 Conversation Engine — must drive both in-app and SMS channels identically. The same check-in or tip dispatches to the right channel based on the patient's account_type. No code-paths that exist only for one track.
- M4 Message Triage (red-flag rule layer first, then NLU; logs versions).
- M5 storage + hot-reload API for the Clinical Content Library, with both full and SMS forms for every message.
- M6 Alerting (fan-out to dashboard, SMS, email; emergency-button webhook for smartphone patients).
- M9 Post-Loss state machine: track activation on clinician mark, content set switch, paced cadence, PHQ-2 dispatch and capture, crisis-keyword route, opt-out handling.
- REST/GraphQL APIs for the dashboard with pagination and access control; consent and audit middleware.

6.3 Data Analyst

- Define the event taxonomy (message_received, message_triage_classified, risk_level_set, check_in_sent, check_in_missed, alert_raised, alert_acknowledged, emergency_button_pressed, appointment_booked, loss_marked, post_loss_opener_sent, phq2_offered, phq2_response, crisis_keyword_detected).
- Stand up the basic dashboards: enrolment funnel by track (smartphone vs choronko), alert volumes by priority, response times, drop-off in conversations, post-loss engagement rate.
- Build the weekly triage-accuracy review queue (samples → Dr Elvira → labels back into M4).
- Track post-loss outcomes separately from pregnancy outcomes — engagement, opt-out rate, crisis-alert frequency.
- Use APHRC IMUA Kenya data and DHS Nigeria/Kenya as background priors for tuning the triage and content.

6.4 Cloud Engineer

- AWS accounts (dev/staging/prod), VPC, networking, IaC in Terraform or AWS CDK.
- Provision Aurora Multi-AZ, Redis, S3 (with KMS), Fargate clusters.
- Cognito user pools, IAM least-privilege, Secrets Manager.
- SMS provider integration (Twilio / Africa's Talking / Termii) for the choronko track — credentials in Secrets Manager, webhook receivers with signature validation.
- Cost monitoring on SMS spend (the choronko track will dominate SMS cost — budget alerts before launch).
- CI/CD pipeline (build, test, security scan, gated deploys).
- CloudWatch dashboards, alarms → on-call; X-Ray tracing.
- Sign the BAA with AWS before go-live; restrict to HIPAA-eligible services.

6.5 Doctor (Dr Elvira, Clinical Lead)

- Provide the patient sign-up clinical questionnaire and the risk-scoring rubric that maps answers to Low / Medium / High risk levels.
- Define the Low / Medium / High message-acuity thresholds and the red-flag keyword list for M4.
- Confirm the wellness check-in cadence per risk level (default: High = daily, Medium = weekly, Low = fortnightly) and define which gestational milestones add extra Low-risk check-ins.
- Author and approve all gestational tips (by week) and all response templates in M5 — non-prescriptive, plain language — in both full and SMS-short forms (a choronko patient must get the same care).
- Approve the patient condition graph thresholds (when does a patient cross from green to yellow to red).

- Lead the design of M9 Post-Loss Care: opening message, paced cadence, the plain-language PHQ-2 wording, the crisis-keyword list, the banned-phrase list ("it's for the best," etc.), and the local crisis-hotline resources.
- Ideally co-author the post-loss content with a perinatal mental-health professional — flag whether one is available or needs to be sourced.
- Review the weekly message-triage accuracy samples from the data analyst and provide corrected labels.

7. MVP Acceptance Criteria

The MVP is done when all of the following are true:

- A hospital can sign up, add personnel, and log in with MFA.
- A smartphone patient can sign up via the app and complete the clinical questionnaire from Dr Elvira.
- A clinician can onboard a choronko (feature-phone) patient at the hospital using only her phone number as identifier. From that moment on, the platform treats her as a full patient.
- A choronko patient receives the same wellness check-ins, gestational tips, and reminders as a smartphone patient — over SMS, at the cadence set by her risk level.
- A choronko patient who texts back a red-flag phrase (e.g. "I am bleeding heavily") raises a High-acuity alert on the dashboard within 30 seconds, exactly as a smartphone patient would.
- At the end of patient sign-up (either track), the system computes and stores her risk level (Low / Medium / High), using Dr Elvira's rubric, and displays it on her dashboard profile.
- Wellness check-ins are sent at the correct cadence for the patient's risk level — daily for High-risk, weekly for Medium-risk, fortnightly for Low-risk.
- When a clinician changes a patient's risk level on the dashboard, the check-in cadence updates within 5 minutes and the change is logged.
- The smartphone patient's home screen shows a working countdown to EDD.
- The bot sends gestational-week tips on schedule, including the practical reminders like "remember to drink water," with the option to also deliver as SMS or as a flash card on the home screen.
- A red-flag message (e.g. "I am bleeding heavily") is classified High acuity, an alert appears on the hospital dashboard within 30 seconds, and the duty clinician is notified by SMS in parallel — regardless of the patient's risk level or track.
- Pressing the emergency button (smartphone only) sends an alert with the patient's GPS to the assigned hospital, visible on the dashboard. For choronko patients, an emergency message reaches the hospital via the same alert pipeline using her registered address.
- The per-patient view on the dashboard shows the condition graph (green/yellow/red) across gestation and marks the patient's appointments and key pregnancy dates.

Post-pregnancy-loss care (M9):

- When a clinician marks a pregnancy as Loss, routine tips, reminders, and the EDD countdown stop within 5 minutes. The post-loss banner appears on her dashboard record.
- Dr Elvira's approved opening message is sent on day 1, and the paced cadence (next message after 48 hours, then settling weekly) is honoured.

- A gentle PHQ-2 check is offered at week 2. The patient's response — or her silence — is recorded on the dashboard without triggering nagging follow-ups.
- A self-harm or crisis keyword in any patient message immediately raises a High alert and triggers a crisis-resource SMS with local hotlines.
- PAUSE / STOP / RESUME keywords are honoured within one message, with a single warm confirmation and no further nudges.
- The post-loss track works identically for smartphone and choronko patients — same content (short-form for SMS), same care, same crisis route.

Privacy and audit:

- Every PHI access, every risk-level change, and every change to triage rules or content is logged in the audit log.

8. Out of Scope for the MVP

- WhatsApp channel (planned for Phase 2). The MVP covers smartphone (app/web) and choronko (SMS).
- Multi-tier referral graph between primary, secondary, and tertiary facilities (Phase 2).
- PHQ-9 and EPDS mental-health screening (Phase 2). The MVP ships only the gentle plain-language PHQ-2 in post-loss care.
- FHIR / EHR export (Phase 2).
- Insurance, billing, and e-prescriptions (the bot will never prescribe — fully excluded).
- Telehealth video calls.

Important — NOT out of scope: post-pregnancy-loss care is part of the MVP (section 2.7 and module M9). It is the most sensitive surface in the product and shipping it well is a launch requirement, not a Phase 2.

9. Risks to Watch

Risk	What we do about it
Message triage misses an emergency.	Red-flag rule layer always runs first, independent of the AI. Conservative tuning. Weekly clinician review of sampled classifications.
Post-loss messaging feels cold or wrong.	Every word reviewed by Dr Elvira (ideally with a perinatal mental-health professional). Banned-phrase list enforced. PAUSE/STOP honoured immediately. Pacing built in — no bombardment after a loss.

Risk	What we do about it
Mental-health crisis missed in the data.	Crisis keywords route to High immediately, hospital alerted, crisis-resource SMS sent in parallel. Dashboard banner makes post-loss status visible to any clinician opening her record.
Choronko patient receives second-class care.	All content authored in both full and SMS forms. M3 dispatches identically on both channels. CI test verifies parity: any tip or check-in that ships to smartphone also has an SMS form.
SMS costs spiral on the choronko track.	Budget alerts on SMS spend. Provider failover. Bulk send windows where appropriate. Daily check-ins for High-risk are the largest cost driver — monitor closely.
Patient on a weak connection misses a tip.	Multi-channel delivery (app push, SMS, flash card) so a single failure doesn't lose the message.
GPS is inaccurate or denied.	App asks for permission with clear reason. If location is unavailable, the alert still goes through with the patient's registered address. Choronko patients always use the registered-address fallback.
Privacy concern.	Encryption at rest and in transit, MFA for staff, full audit log, least-privilege access, BAA with AWS.

— End of MVP SRS —